Hazard Analysis
Lake

Team #10, CinqGarçons
Ethan Walsh
Colin Chambachan
Matthew Nesbitt
Jinwoo Hong
Gary Qin

Table 1: Revision History

Date	**Developer(s)**	**Change**
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

Contents

1 Introduction **1**

2 Scope and Purpose of Hazard Analysis **1**

3 System Boundaries and Components **1**
- 3.1 Coordinator . . . 1
- 3.2 Model Partitioner . . . 1
- 3.3 Inference Engine . . . 1
- 3.4 Tensor I/O . . . 2
- 3.5 Boundaries . . . 2

4 Critical Assumptions **2**

5 Failure Mode and Effect Analysis **2**

6 Safety and Security Requirements **2**

7 Roadmap **3**

8 Appendix — Generative AI Use Disclosure **4**
- 8.1 AI Tools Used . . . 4
- 8.2 How and Where Used . . . 4
- 8.3 Verification of AI-Generated Content . . . 4

9 Appendix — Reflection **5**

[You are free to modify this template. —SS]

1 Introduction

[You can include your definition of what a hazard is here. —SS]

2 Scope and Purpose of Hazard Analysis

[You should say what **loss** could be incurred because of the hazards. —SS]

3 System Boundaries and Components

3.1 Coordinator

This component exists on all nodes, but the following responsibilities are only run on the leader node:

- Providing an OpenAI-API compatible API to the user for use with tools like OpenCode.
- Tracking cluster health, and detecting node failures.
- Notifying nodes of partitioning changes.
- Rebuilding the cluster and saving in-progress requests in the case of node failure, removal, or addition.
- Measuring node connection performance.

On a follower node, it is primarily responsible for managing the inference engine on its device, following the instructions of the leader node.

3.2 Model Partitioner

This component determines how to partition the model among the participating devices. It decides at which points to cut the graph, how large each partition is, and which device gets each partition. This component is integral to ensuring that each device is assigned an appropriate partition given the device's computational and networking performance.

3.3 Inference Engine

This component is the high-performance inference engine running on each device. It will handle execution of the given model partition for each device. It will call into ggml using the Rust-to-C FFI (TODO: glossary) to control the execution of a device's model partition.

3.4 Tensor I/O

This component handles the compressing, sending, receiving, and decompressing of tensors between devices. It will leverage gRPC for high-performance communication, with the potential to use a custom solution should gRPC fail to meet our requirements in practice.

3.5 Boundaries

The system boundaries for Lake primarily consist of:

- The Coordinator's OpenAI-API compatible API.
- Model files (GGUF) loaded from disk.
- Networking components between devices (Thunderbolt, Ethernet, Wi-Fi, routers).
- The OS and GPU resources on each individual device.

4 Critical Assumptions

We assume:

- The model file (GGUF) provided to Lake is already a functional LLM before partitioning.
- User requests are only made through the API on the leader device.
- Only one model will be run by the pool at a time.
- The local area network connecting the devices is private, and trusted by the user.

5 Failure Mode and Effect Analysis

[Include your FMEA table here. This is the most important part of this document. —SS] [The safety requirements in the table do not have to have the prefix SR. The most important thing is to show traceability to your SRS. You might trace to requirements you have already written, or you might need to add new requirements. —SS] [If no safety requirement can be devised, other mitigation strategies can be entered in the table, including strategies involving providing additional documentation, and/or test cases. —SS]

6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

8 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

8.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

8.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

8.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

9 Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?